University of Toronto

Master of Mathematical Finance

Treasury and Liquidity Risk

Assignment – Fiscal Q3 2026

Risk Management Module

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Scope and methodology

This assignment compares the consolidated results of Royal Bank of Canada (RBC), Toronto-Dominion Bank (TD), Bank of Montreal (BMO), Bank of Nova Scotia (Scotiabank), and Canadian Imperial Bank of Commerce (CIBC), using their fiscal Q3 2026 Reports to Shareholders [1, 2, 3, 4, 5].

Q3 refers to the three months ended July 31, 2026, and Q2 to the three months ended April 30, 2026. Year-to-date (YTD) results cover the first nine months of the relevant fiscal year.

Unless otherwise noted, monetary amounts are in **CAD millions**. A change in a ratio expressed in percentage terms is reported in *percentage points* (pp); a change in a balance is reported as a percentage of its Q2 balance.

Amounts in parentheses in the source reports are treated as negative. For BMO's liquidity tables, amounts originally reported in CAD billions have been converted to CAD millions.

Comparability note. Scotiabank reports its Q3 2026 LCR and NSFR under OSFI's revised May 2026 Liquidity Adequacy Requirements (LAR) Guideline, whereas Q2 2026 figures were reported under the April 2025 guideline.

Therefore, changes in these ratios may partly reflect differences in regulatory methodology, in addition to changes in the bank's liquidity and funding positions.

1 Net Interest Income

Net Interest Income (NII) is interest earned on assets such as loans and securities, less interest paid on liabilities such as deposits and wholesale funding. It is distinct from net income, which also reflects fees, operating expenses, provisions for credit losses and taxes.

Table 1: Consolidated quarterly NII (CAD millions)

Bank	Q2 2026	Q3 2026	Change	Change (%)
RBC	8,506	8,744	+238	+2.8
TD	8,861	9,296	+435	+4.9
BMO	5,268	5,567	+299	+5.7
Scotiabank	5,521	5,866	+345	+6.2
CIBC	4,345	4,507	+162	+3.7

Table 2: Consolidated year-to-date NII (CAD millions)

Bank	YTD Q3 2025	YTD Q3 2026	Change	Change (%)
RBC	24,355	25,835	+1,480	+6.1
TD	24,517	26,946	+2,429	+9.9
BMO	15,991	16,478	+487	+3.0
Scotiabank	15,936	16,969	+1,033	+6.5
CIBC	11,637	13,160	+1,523	+13.1

The percentage change is computed as

$$\%\Delta\text{NII} = 100\left(\frac{\text{NII}_{\text{new}}}{\text{NII}_{\text{old}}} - 1\right).$$

All five banks increased their consolidated NII quarter over quarter and on a YTD basis. The quarterly increases have different underlying drivers:

RBC. NII increased by CAD 238 million (2.8%). RBC attributes this mainly to three additional days in Q3 and growth in average loan and other interest-earning balances in Personal Banking and Commercial Banking.

More days allow interest to accrue for longer, while higher balances generate additional interest income. This is not evidence that the bank's interest margin necessarily improved: its reported consolidated net interest margin actually fell compared with Q2 [1, p. 8].

TD. NII increased by CAD 435 million (4.9%). TD explicitly highlights the additional days in the quarter and the favourable effect of translating foreign-currency results into Canadian dollars.

The increase was concentrated in Canadian Personal and Commercial Banking and U.S. Banking, with smaller contributions from Wealth Management and Insurance and Corporate; Wholesale Banking slightly offset these gains [2, p. 12].

BMO. NII increased by CAD 299 million (5.7%). BMO identifies three additional days, balance growth across businesses, and higher NII in Global Markets as the principal drivers.

Lower NII in Corporate Services partially offset these factors. The distinction matters because an increase in total NII can arise from a larger volume of business even when the overall interest margin is flat or slightly lower [3, p. 14].

Scotiabank. NII increased by CAD 345 million (6.2%), mainly because of loan growth, higher net interest income from capital-markets activities, and three additional days in the quarter.

Its consolidated net interest margin remained at 2.49%: margin improvements in some business lines were offset by weaker margins in International Banking.

The growth in NII was therefore not primarily a bank-wide margin expansion story [4, p. 20].

CIBC. NII increased by CAD 162 million (3.7%), reflecting the longer quarter, higher business volumes, stronger margins in non-trading businesses, and higher Treasury revenue.

Lower trading NII offset part of the increase. This illustrates why it is important to use consolidated NII rather than just non-trading NII: both trading and non-trading activities contribute to the reported total [5, pp. 4–5].

For the YTD comparison, CIBC recorded the fastest percentage growth (13.1%), while BMO recorded the slowest (3.0%).

The YTD increases should also be interpreted in light of changes in business mix: for instance, Scotiabank reports an offsetting effect from divested operations [4, p. 20].

2 NII sensitivity to a +100 bps shock

NII sensitivity estimates how much the bank's projected net interest income over the next 12 months would change following an immediate and sustained 100-basis-point (one-percentage-point) increase in interest rates.

The sensitivities below are *pre-tax changes*, not the levels of NII reported in Question 1.

Table 3: 12-month NII sensitivity to +100 bps (CAD millions)

Bank	**Q2 2026**	**Q3 2026**	**Change**
RBC	+187	+373	+186
TD	+411	+341	-70
BMO	+233	+291	+58
Scotiabank	+197	+222	+25
CIBC	+119	+177	+58

All five banks have *positive* NII sensitivity: under the specified scenario, their projected NII would increase when rates rise.

This can occur when the return earned on rate-sensitive assets adjusts more quickly, or by more, than the interest paid on funding. Deposits with low rate sensitivity ("low deposit beta") can contribute to this outcome.

RBC experienced the largest increase in its positive sensitivity, from CAD 187 million to CAD 373 million. BMO and CIBC also became more positively sensitive, while Scotiabank's sensitivity increased modestly.

TD moved in the opposite direction: its estimated benefit from rising rates declined by CAD 70 million, although it remained positive.

TD links its quarterly changes in structural interest-rate sensitivities mainly to Treasury hedging activity; RBC refers to a reduction in its fixed-rate net asset position, including model-parameter updates [1, p. 29][2, p. 36].

These are scenario estimates, not forecasts of next quarter's actual NII. Banks' balance sheets, hedges and customer behaviour can change, and methodological details are not necessarily identical across banks.

BMO labels its disclosed measure *12-month earnings sensitivity* within structural interest-rate risk; it is used here as the closest disclosed counterpart to the requested NII sensitivity [3, p. 34].

3 EVE sensitivity to a +100 bps shock

Economic Value of Equity (EVE) sensitivity measures the change in the estimated present value of banking-book assets and relevant off-balance-sheet positions, less the present value of liabilities, following the same interest-rate shock.

Unlike 12-month NII sensitivity, EVE looks at the value of cash flows over their relevant remaining lives.

All five banks report *negative* EVE sensitivity to a rise of 100 bps. In other words, the immediate rate increase reduces the estimated economic value of their net banking-book positions.

The change in sensitivity must be interpreted with its sign. RBC's EVE impact improved from -2,711 to -2,511 million: the hypothetical loss became CAD 200 million smaller.

Scotiabank also became slightly less exposed to this scenario, improving by CAD 35 million.

In contrast, TD's hypothetical loss increased by CAD 123 million, BMO's by CAD 51 million and CIBC's by CAD 13 million.

Table 4: EVE sensitivity to +100 bps (CAD millions)

Bank	Q2 2026	Q3 2026	Change
RBC	-2,711	-2,511	+200
TD	-3,683	-3,806	-123
BMO	-2,059	-2,110	-51
Scotiabank	-1,871	-1,836	+35
CIBC	-1,542	-1,555	-13

For RBC, the reported improvement is consistent with the bank's explanation of a lower fixed-rate net asset position; for BMO, the bank discusses greater modelled deposit-pricing sensitivity following higher term rates.

These interpretations refer to the banks' own disclosures, rather than assuming that every change came from the same source [1, p. 29][3, p. 34].

A larger negative number does not, by itself, prove that one bank is less prudently managed than another: the banks differ in size, business mix, customer behaviour, hedging strategy and modelling assumptions.

4 Why can NII and EVE sensitivities have opposite signs?

There is no contradiction because the two measures answer different questions.

NII sensitivity asks how much the bank expects to earn in net interest over the *next 12 months*.

EVE sensitivity asks how the *present value of future net cash flows* changes immediately when the discount curve shifts.

Consider a bank with many long-term fixed-rate mortgages funded partly by shorter-term deposits, as well as some floating-rate assets.

When rates rise, interest income from floating-rate assets and newly repriced loans can increase quickly, while the cost of sticky deposits may increase by less. This can produce

$$\Delta \text{NII} > 0$$

over the next year.

At the same time, the future cash flows on existing fixed-rate mortgages remain largely unchanged. Discounting them at higher rates reduces their present value.

Although the present value of some liabilities may also decline, the bank's assets can have greater effective duration than its liabilities, resulting in

$$\Delta \text{EVE} < 0.$$

In simplified terms,

$$\text{EVE} = PV(\text{assets}) - PV(\text{liabilities}),$$

$$\Delta\text{EVE} \approx -(D_A A - D_L L)\,\Delta y,$$

where D_A and D_L are effective durations, A and L are present values, and Δy is the rate shock.

The duration expression is an approximation, not a substitute for the banks' behavioural models.

Thus a rate increase may *improve near-term interest earnings while reducing the present value of longer-term net assets.*

This is exactly the sign pattern observed in Q3 2026 for all five banks.

Repricing mismatches, deposit betas, embedded options and interest-rate hedges determine the size of the two effects.

5 More floating-rate mortgages: implications for NII and EVE

If customers increasingly choose floating-rate rather than fixed-rate mortgages, the bank's asset portfolio generally becomes quicker to reprice.

A floating mortgage's coupon adjusts when its reference rate changes, while a fixed-rate mortgage continues to pay its agreed coupon until repricing or maturity.

NII sensitivity.

With more floating-rate mortgages, a +100 bps shock would generally raise mortgage interest income sooner.

If the deposits funding those loans reprice more slowly, the bank's net interest margin benefits, making NII sensitivity to rising rates more positive.

However, the result depends on the funding mix: if mortgage funding costs also reprice quickly, the additional benefit may be small.

Hedging can change it further.

EVE sensitivity.

Floating-rate mortgages generally have lower interest-rate duration than otherwise comparable fixed-rate mortgages because their coupons reset toward market rates.

Replacing fixed-rate mortgages with floating-rate ones would therefore tend to reduce the loss of asset value when rates rise.

Other things equal, a bank with negative EVE sensitivity might see that sensitivity become *less negative.*

These are directional expectations, not mechanical predictions.

The ultimate impact depends on whether the mortgages replace existing fixed-rate assets, the effective maturity of the deposits, mortgage prepayment behaviour, and the bank's use of swaps and other hedges.

A floating-rate mortgage portfolio does not automatically imply zero interest-rate risk.

6 Liquidity Coverage Ratio (LCR)

The LCR evaluates whether a bank holds enough High-Quality Liquid Assets (HQLA) to cover estimated net cash outflows during a severe 30-day liquidity stress:

$$\text{LCR} = \frac{\text{HQLA}}{\text{30-day net stressed cash outflows}} \times 100\%.$$

The regulatory minimum for these banks is normally 100%, so a ratio above 100% indicates a buffer above the minimum under the regulatory scenario.

Table 5: Reported LCR (percent)

Bank	Q2 2026	Q3 2026	Change (pp)
RBC	126%	125%	-1
TD	130%	133%	+3
BMO	128%	125%	-3
Scotiabank	124%	126%	+2
CIBC	131%	127%	-4

All five banks remained above the 100% regulatory threshold.

TD and Scotiabank increased their LCR, while RBC, BMO and CIBC reported declines.

The changes are best understood by separately examining HQLA and net cash outflows, rather than interpreting a lower ratio as automatically indicating an absolute reduction in liquid assets.

7 What drives the quarterly change in LCR?

Table 6: LCR components (CAD millions; changes relative to Q2)

	HQLA (numerator)			Net cash outflows (denominator)		
Bank	**Q2**	**Q3**	**Change (%)**	**Q2**	**Q3**	**Change (%)**
RBC	469,804	486,851	+3.63	373,537	388,550	+4.02
TD	336,237	333,143	-0.92	258,225	250,709	-2.91
BMO	256,200	257,400	+0.47	200,600	206,000	+2.69
Scotiabank	290,089	311,181	+7.27	233,382	246,974	+5.82
CIBC	204,468	211,128	+3.26	155,702	166,647	+7.03

RBC: mainly higher net cash outflows.

HQLA rose by 3.63%, but net cash outflows rose faster, by 4.02%.

The denominator therefore slightly outpaced the buffer and the LCR declined from 126% to 125%.

The decline *does not* mean RBC held fewer liquid assets.

TD: mainly lower net cash outflows.

HQLA fell by 0.92%, but net cash outflows fell more sharply, by 2.91%.

Because the denominator shrank faster than the numerator, the LCR improved from 130% to 133%.

BMO: mainly higher net cash outflows.

HQLA increased only 0.47%, while net cash outflows increased 2.69%.

The stronger rise in the denominator explains why the LCR declined from 128% to 125%, consistent with BMO's own explanation [3, p. 38].

Scotiabank: mainly higher HQLA.

The stock of HQLA grew 7.27%, more than the 5.82% increase in net cash outflows.

This increased the reported LCR from 124% to 126%.

The bank likewise identifies higher HQLA as the main factor, partly offset by larger outflows [4, p. 48].

CIBC: mainly higher net cash outflows.

HQLA increased 3.26%, but outflows rose 7.03%, leading to a decline in LCR from 131% to 127%.

CIBC identifies higher secured wholesale funding, less stable deposits and non-operational deposits among the sources of the increased outflows [5, p. 41].

Measurement note.

LCR figures are reported as averages of daily ratios during the quarter.

The ratio of the disclosed average HQLA to the disclosed average outflows need not exactly equal the average daily LCR.

Comparisons of the components above explain the direction of the movement, without claiming an exact arithmetic decomposition of the published ratio.

8 Net Stable Funding Ratio (NSFR)

The NSFR compares Available Stable Funding (ASF), such as regulatory capital, stable deposits and longer-term borrowing, with Required Stable Funding (RSF) for the bank's assets and off-balance-sheet commitments, over a one-year horizon:

$$\text{NSFR} = \frac{\text{ASF}}{\text{RSF}} \times 100\%.$$

A ratio above 100% indicates that the bank has more regulatory stable funding available than required.

Again, all banks are above the regulatory minimum of 100%.

RBC's reported NSFR rose, TD's and CIBC's fell, and BMO's and Scotiabank's ratios were unchanged after rounding to whole percentages.

To identify the driver of these movements, we must examine ASF and RSF separately.

Table 7: Reported NSFR (percent)

Bank	Q2 2026	Q3 2026	Change (pp)
RBC	111%	112%	+1
TD	117%	114%	-3
BMO	114%	114%	0
Scotiabank	116%	116%	0
CIBC	114%	113%	-1

9 What drives the quarterly change in NSFR?

Table 8: NSFR components (CAD millions; changes relative to Q2)

	ASF (numerator)			RSF (denominator)		
Bank	**Q2**	**Q3**	**Change (%)**	**Q2**	**Q3**	**Change (%)**
RBC	1,181,392	1,225,575	+3.74	1,066,526	1,091,878	+2.38
TD	1,171,411	1,189,463	+1.54	1,004,256	1,044,384	+4.00
BMO	794,400	816,700	+2.81	699,200	719,300	+2.87
Scotiabank	812,791	837,075	+2.99	700,994	720,128	+2.73
CIBC	645,887	659,366	+2.09	565,530	582,482	+3.00

RBC: ASF growth dominates.

ASF increased 3.74%, faster than the 2.38% rise in RSF.

The stronger increase in stable funding relative to funding needs explains the improvement in NSFR from 111% to 112%.

TD: RSF growth dominates.

ASF increased 1.54%, but RSF increased 4.00%.

The larger growth in funding required for assets and exposures explains the decline in NSFR from 117% to 114%.

TD therefore remains compliant but has a smaller stable-funding cushion in ratio terms.

BMO: almost exactly offsetting changes.

ASF increased 2.81% and RSF 2.87%.

These nearly proportional increases explain the unchanged reported NSFR of 114%.

It would be misleading to assign the movement to only one side: both stable funding and funding requirements expanded at almost the same rate [3, p. 39].

Scotiabank: broadly offsetting changes.

ASF increased 2.99%, compared with a 2.73% increase in RSF.

The small difference was not enough to change the rounded reported NSFR of 116%.

In unrounded terms, the simple ASF/RSF quotient increases slightly [4, pp. 48–49].

CIBC: RSF growth dominates.

ASF increased 2.09%, but RSF rose 3.00%.

CIBC specifically relates higher RSF to performing loans and securities, partly offset by additional ASF from wholesale funding.

This accounts for the decline in NSFR from 114% to 113% [5, p. 42].

Conclusion

The five banks reported higher consolidated NII in Q3, but their interest-rate and liquidity sensitivities did not all move in the same direction.

The contrast between positive NII sensitivity and negative EVE sensitivity highlights the difference between short-term earnings and long-term economic value.

Similarly, changes in LCR and NSFR are only meaningful when their numerators and denominators are considered together.

Although all five banks remained above the regulatory LCR and NSFR minima, their quarter-on-quarter movements reflect different changes in liquidity buffers, stressed outflows, available funding and required funding.

References

[1] Royal Bank of Canada, *Third Quarter 2026 Report to Shareholders*, pp. 8 (NII and quarterly drivers), 29 (interest-rate sensitivity), 37 (LCR), and 39 (NSFR).

[2] TD Bank Group, *Third Quarter 2026 Report to Shareholders*, pp. 8 (NII), 12 (drivers), 36 (interest-rate sensitivity), 41 (LCR), and 42 (NSFR).

[3] BMO Financial Group, *Third Quarter 2026 Report to Shareholders*, pp. 7 (NII), 14 (drivers), 34 (interest-rate sensitivity), 38 (LCR), and 39 (NSFR).

[4] Scotiabank, *Q3 2026 Quarterly Report to Shareholders*, pp. 18 (NII), 20 (drivers), 44 (interest-rate sensitivity), 48 (LCR), and 49 (NSFR).

[5] Canadian Imperial Bank of Commerce, *Report to Shareholders for the Third Quarter, 2026*, pp. 4–5 (NII and drivers), 38 (interest-rate sensitivity), 41 (LCR), and 42 (NSFR).